

IRWIN AND JOAN JACOBS
CENTER FOR COMMUNICATION AND INFORMATION TECHNOLOGIES

Subset-sum phase transitions and data compression

Neri Merhav

CCIT Report #792
July 2011



DEPARTMENT OF ELECTRICAL ENGINEERING
TECHNION - ISRAEL INSTITUTE OF TECHNOLOGY, HAIFA 32000, ISRAEL



CCIT Report #792 July 2011

Subset–sum phase transitions and data compression

Neri Merhav

Department of Electrical Engineering, Technion, Haifa 32000, Israel.
E–mail: merhav@ee.technion.ac.il

Abstract. We propose a rigorous analysis approach for the subset sum problem in the context of lossless data compression, where the phase transition of the subset sum problem is directly related to the passage between ambiguous and non–ambiguous decompression, for a compression scheme that is based on specifying the sequence composition. The proposed analysis lends itself to straightforward extensions in several directions of interest, including non–binary alphabets, incorporation of side information at the decoder (Slepian–Wolf coding), and coding schemes based on multiple subset sums. It is also demonstrated that the proposed technique can be used to analyze the critical behavior in a more involved situation where the sequence composition is not specified by the encoder.

Keywords: Number partitioning, integer partitioning, subset sum, data compression, source coding, entropy, phase transitions.

1. Introduction

We consider a lossless data compression scheme that builds upon the the *number partitioning* problem and the closely related problem of *subset sums*: Given a set of integers, $a_1, a_2, \ldots, a_N$, $a_i \in \{1, 2, \ldots, L\}$, $i = 1, 2, \ldots, N$, the number partitioning problem is the problem of finding a subset $\mathcal{S} \subseteq \{1, 2, \ldots, N\}$, such that the sums of $\{a_i\}$ over $\mathcal{S}$ would be as balanced as possible with the sum over the remaining $\{a_i\}$. More precisely, the goal is to find a subset $\mathcal{S}$ such that $|\sum_{i\in\mathcal{S}} a_i - \sum_{i\in\mathcal{S}^c} a_i|$ would be minimum, or equivalently, to find a binary vector $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_N) \in \{-1, +1\}^N$ such that $|\sum_{i=1}^N a_i\sigma_i|$ would be minimum. Perfect partitioning means that this expression is exactly equal to zero. The problem of finding an optimum partition is NP–complete [6], [11] and it has a fairly long history (see, e.g., [10, Section 9.2], [9, Chapter 7] and many references therein). For the case where $\{a_i\}$ are drawn independently at random, some rigorous results have been obtained using methods of statistical mechanics, see, e.g, [1], [2], [5], [8]. It has been shown (see also [9], [10], [13]) that for a randomly selected vector $(a_1, \ldots, a_N)$, and for $L = 2^{NR}$ ($R > 0$, constant), there is a phase transition at $R = 1$. For $R < 1$, there are exponentially many solutions (σ–vectors) to the number partitioning problem. More precisely, there are exponentially about $2^{N(1-R)}$ many solutions on the average. However, for $R > 1$, the probability that there exists even one solution decays exponentially.

In the related problem of subset sums, the scope is extended to the evaluation of the total number $\Omega(E)$ of binary vectors $\{\boldsymbol{\sigma}\}$ such that $\sum_{i=1}^N a_i\sigma_i = E$, for any given value of E in the appropriate range, not only $E = 0$. Sasamoto [12] proposed a data compression scheme based on subset sums in its constrained version, that is, the one where binary vectors are sought only among those which have a given *composition*, namely, given numbers $N_+ = Np$ ($0 \le p \le 1$) and $N_- = Nq$ ($q = 1-p$) of occurrences of $\sigma_i = +1$ and $\sigma_i = -1$, respectively, or equivalently, a given value of $M(\boldsymbol{\sigma}) = \sum_{i=1}^N \sigma_i = N(p-q)$.‡ In particular, in view of the above described results concerning phase transitions, Sasamoto's insight was that for R above a certain threshold, the mapping between the set of binary vectors $\{\boldsymbol{\sigma}\}$ of a given composition to the sums $E(\boldsymbol{\sigma}) = \sum_{i=1}^N a_i\sigma_i$ must be essentially one–to-one for a typical realization of $(a_1, \ldots, a_N)$. This has lead him to propose a lossless data compression scheme that is based on encoding a binary string $\boldsymbol{\sigma}$, with a composition of $N_+ = Np$ and $N_- = Nq$, using a binary representation of $E(\boldsymbol{\sigma})$ plus a relatively small overhead (of $\log(N+1)$ bits) for specifying the composition of $\boldsymbol{\sigma}$, or equivalently, the value of $M(\boldsymbol{\sigma}) = \sum_{i=1}^N \sigma_i$. Sasamoto argued that the threshold of reliable decoding occurs at $R = h(p)$, where

$$h(p) = -p\log p - (1-p)\log(1-p) \tag{1}$$

is entropy of the binary information source that emits sequences with the aforementioned composition (within some small tolerance) with high probability, and so by taking $R = h(p)+\epsilon$ ($\epsilon > 0$, arbitrarily small) and using the fact that the range of possible values

‡ By contrast, in the unconstrained version, solutions are sought across all binary strings of length N.

of $E(\boldsymbol{\sigma})$ does not exceed $N \cdot L$, one may encode $E(\boldsymbol{\sigma})$ using $\log(N \cdot L) = N[h(p)+\epsilon]+\log N$ bits, and thereby essentially achieve the entropy of the information source. While this coding scheme is not very attractive from the practical point of view, the interesting point here is the relationship between the phase transition of the subset problem and the abrupt passage between ambiguous and non–ambiguous decoding as R crosses the entropy, in agreement with Shannon's fundamental coding theorems [3].

Sasamoto's approach was to analyze the number $\Omega(E, M)$ of configurations $\{\boldsymbol{\sigma}\}$ with $E(\boldsymbol{\sigma}) = E$ and $M(\boldsymbol{\sigma}) = M$, where E and M are the values pertaining to the source sequence $\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_1, \ldots, \hat{\sigma}_N)$ that was actually compressed. He argued that for a typical realization of $\{a_i\}$, the behavior is as follows: For $R < h(p)$, $\Omega(E, M)$ is exponentially large and so the decoding of $\hat{\boldsymbol{\sigma}}$, based on E and M, is ambiguous, but for $R > h(p)$, the expectation of $\Omega(E, M)$ is exponentially small, and so the decoding is reliable with high probability. In order to assess the number of solutions to the two simultaneous equations $E(\boldsymbol{\sigma}) = E$ and $M(\boldsymbol{\sigma}) = M$, he applied the saddle point method (see also [10], [13]). In particular, he first defined a partition function of a Hamiltonian defined by a linear combination of $E(\boldsymbol{\sigma})$ and $M(\boldsymbol{\sigma})$, and then used the integral representation of the inverse transform of this partition function, that yields $\Omega(E, M)$. This integral in turn was approximated using the saddle point method.

The analysis in [12], which relies on the analysis in [13], raises two technical concerns, however. The first is about the validity of the saddle point method in this situation: While the saddle point method is perfectly rigorous under the asymptotic regime where $N \to \infty$ while L is kept fixed, its validity becomes rather questionable§ in a regime where L grows with N, especially when the growth rate of L is as fast as exponential. The authors of [13] realize that the resulting approximation is definitely not valid when $R > 1$, which yields $\Omega(E, M) < 1$. The point, however, is that it is not quite clear whether this approximation is reliable even when $R < 1$. The fact that the resulting approximation below $R = 1$ does not lead to an obvious absurd is not enough to guarantee that the approximation is reliable.

The second concern is that there is a difference between calculating the expectation of $\Omega(E, M)$ when E and M are fixed and deterministic, and calculating the expectation of $\Omega(E, M)$ when $M = M(\hat{\boldsymbol{\sigma}})$ and $E = E(\hat{\boldsymbol{\sigma}}) = \sum_i a_i \hat{\sigma}_i$, because the latter is a *random variable*. The former quantity is what Sasamoto calculated and the latter is actually the relevant quantity for analyzing the data compression scheme. When computing the expectation of $\Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}}))$, the randomness of $E(\hat{\boldsymbol{\sigma}})$ is induced by the same set of random variables $\{a_i\}$ that generate also the values of $E(\boldsymbol{\sigma})$ pertaining to all other binary vectors $\{\boldsymbol{\sigma}\}$. In other words, in this calculation both the function $\Omega(\cdot, \cdot)$ and its first argument $E(\hat{\boldsymbol{\sigma}})$ fluctuate together, depending on $\{a_i\}$. Indeed, for one thing, $\Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}}))$ (and hence also its expectation) must always be at least as large as unity (by construction), whereas the expectation of $\Omega(E, M)$ for fixed E and M is shown in [12] to decay exponentially to zero for R above the threshold. This is clearly

§ In [13] there are detailed discussions about the validity of the saddle point method when L is a function of N (see the ending paragraph of Section 3 on page 9559 and pp. 9563–9564 therein).

a contradiction.

In this work, we first propose a rigorous approach to evaluate the expectation of $\Omega(E(\hat{\boldsymbol{\sigma}}, M(\hat{\boldsymbol{\sigma}}))$, which is valid for every $R \geq 0$. Our starting point is (or can be interpreted as) essentially the same inverse transform integral of the above–mentioned partition function (but with a slight modification to account for the above discussed replacement of a fixed E by $E(\hat{\boldsymbol{\sigma}})$). However, unlike in [12] and [13], we avoid the use of the saddle point method in the evaluation of this integral and we propose a more refined analysis instead. The final result of this analysis is similar to that of [12] for R below the threshold, but it is not quite identical above the threshold: We show that when the relative frequencies of $+1$ and -1 in $\hat{\boldsymbol{\sigma}}$ are p and q, respectively, and $L = 2^{NR}$,

$$\langle \Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}}))\rangle \doteq 1 + 2^{N[h(p)-R]}, \tag{2}$$

where $\langle\cdot\rangle$ denotes expectation w.r.t. the randomness of $\{a_i\}$ and $\doteq$ means equality in the exponential order sense ($a_N \doteq b_N$ means that $\frac{1}{N}\ln\frac{a_N}{b_N} \to 0$ as $N \to \infty$). Thus, indeed there is a phase transition at $R = h(p)$: For $R < h(p)$, there are exponentially many source vectors that are mapped to the same value of $E(\hat{\boldsymbol{\sigma}})$ on the average, but for $R > h(p)$ the expected number of additional source vectors (other than $\hat{\boldsymbol{\sigma}}$) vanishes. Since $\Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}}))$ is an integer–valued random variable, this also means (by the Chebychev inequality) that $\Pr\{\Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}})) > 1\}$ also vanishes for $R > h(p)$.

While the final conclusions of our analysis are essentially the same as in [12] (for $R < h(p)$), the message in this paper is three–fold: The first message is that it is not necessary to resort to the saddle point method in this case and it is possible to make the analysis rigorous as we show. The second message is that our analysis extends easily to more general situations, like larger source alphabets, availability of side information at the decoder (a.k.a. Slepian–Wolf encoding [3, Section 15.8], [14]), and so on. The third message is that this analysis method can be used also to handle non–trivial alternative coding schemes, like a scheme based on the unconstrained subset sum problem. It turns out that for such a scheme, the phase transition occurs at a critical value of R which is different from the entropy $h(p)$, and we provide an explicit expression, which is not trivial.

2. Constrained Subset–Sum Coding

Consider first the problem of counting the number of solutions $\{\boldsymbol{\sigma}\}$ to the two simultaneous equations

$$E(\boldsymbol{\sigma}) = E(\hat{\boldsymbol{\sigma}}) \tag{3}$$

$$M(\boldsymbol{\sigma}) = M(\hat{\boldsymbol{\sigma}}) \tag{4}$$

Denoting the Kronecker Delta function by $\delta(\cdot)$ and $\sqrt{-1}$ by i, we have

$$\begin{aligned}\Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}})) &= \sum_{\boldsymbol{\sigma}} \delta(E(\hat{\boldsymbol{\sigma}}) - E(\boldsymbol{\sigma}))\delta(M(\hat{\boldsymbol{\sigma}}) - M(\boldsymbol{\sigma})) \\ &= \sum_{\boldsymbol{\sigma}} \delta\left(\sum_j a_j(\hat{\sigma}_j - \sigma_j)\right)\delta\left(N_+ - N_- - \sum_j \sigma_j\right)\end{aligned}$$

$$
\begin{aligned}
&= \sum_{\boldsymbol{\sigma}} \int_{-\pi}^{+\pi} \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega \mathrm{d}\theta}{(2\pi)^2} \exp\left\{ i\omega \left[\sum_j a_j(\hat{\sigma}_j - \sigma_j) \right] \right\} \times \\
&\quad \exp\left\{ i\theta \left(N_+ - N_- - \sum_j \sigma_j) \right) \right\} \\
&= \int_{-\pi}^{+\pi} \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega \mathrm{d}\theta}{(2\pi)^2} e^{i\theta(N_+ - N_-)} \times \\
&\quad \prod_{j=1}^{N} \sum_{\sigma_j=-1}^{+1} \exp\{ i\omega a_j(\hat{\sigma}_j - \sigma_j) - i\theta\sigma_j \} \\
&= \int_{-\pi}^{+\pi} \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega \mathrm{d}\theta}{(2\pi)^2} e^{i\theta(N_+ - N_-)} \prod_{j=1}^{N} \left[e^{-i\theta\hat{\sigma}_j} + e^{i(\theta + 2\omega a_j)\hat{\sigma}_j} \right] \\
&= \int_{-\pi}^{+\pi} \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega \mathrm{d}\theta}{(2\pi)^2} e^{i\theta(N_+ - N_-)} \prod_{j:\ \hat{\sigma}_j=+1} \left[e^{-i\theta} + e^{i(\theta + 2\omega a_j)} \right] \times \\
&\quad \prod_{j:\ \hat{\sigma}_j=-1} \left[e^{i\theta} + e^{-i(\theta + 2\omega a_j)} \right] \\
&= \int_{-\pi}^{+\pi} \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega \mathrm{d}\theta}{(2\pi)^2} \cdot \prod_{j:\ \hat{\sigma}_j=+1} \left[1 + e^{2i(\theta + \omega a_j)} \right] \times \\
&\quad \prod_{j:\ \hat{\sigma}_j=-1} \left[1 + e^{-2i(\theta + \omega a_j)} \right]. \qquad (5)
\end{aligned}
$$

Taking now the expectation over $\{a_i\}$, and denoting

$$
\phi(\omega) = \left\langle e^{2i\omega a_1} \right\rangle = \frac{1}{L} \sum_{j=1}^{L} e^{2i\omega j}, \qquad (6)
$$

we have:

$$
\begin{aligned}
\langle \Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}})) \rangle &= \int_{-\pi}^{+\pi} \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega \mathrm{d}\theta}{(2\pi)^2} \left[1 + e^{2i\theta} \phi(\omega) \right]^{N_+} \times \\
&\quad \left[1 + e^{-2i\theta} \phi(-\omega) \right]^{N_-} \\
&= 1 + \sum_n \sum_k \binom{N_+}{n} \binom{N_-}{k} \times \\
&\quad \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \phi^n(\omega) \phi^k(-\omega) \int_{-\pi}^{+\pi} \frac{\mathrm{d}\theta}{2\pi} e^{2i\theta(n-k)} \\
&= 1 + \sum_{n=1}^{\min\{N_+, N_-\}} \binom{N_+}{n} \binom{N_-}{n} \times \\
&\quad \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} [\phi(\omega)\phi(-\omega)]^n \\
&= 1 + \sum_{n=1}^{\min\{N_+, N_-\}} L^{-2n} \binom{N_+}{n} \binom{N_-}{n} \sum_{s=n}^{nL} (\Lambda_s^n)^2, \qquad (7)
\end{aligned}
$$

where the summation over n and k in the second line is over $\{0, 1, \ldots, N_+\} \times \{0, 1, \ldots, N_-\} \setminus \{0, 0\}$ and where Λ_s^n is the number of vectors $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_n) \in \{1, 2, \ldots, L\}^n$ with $\sum_{i=1}^n \alpha_i = s$. The last line of eq. (7) has a simple interpretation:

Let $\mathcal{N}_+ = \{i : \sigma_i = +1\}$, $\mathcal{N}_- = \{i : \sigma_i = -1\}$, $\hat{\mathcal{N}}_+ = \{i : \hat{\sigma}_i = +1\}$, and $\hat{\mathcal{N}}_- = \{i : \hat{\sigma}_i = -1\}$. Obviously, $E(\boldsymbol{\sigma}) = E(\hat{\boldsymbol{\sigma}})$ if and only if

$$\sum_{i \in \mathcal{N}_+ \cap \hat{\mathcal{N}}_-} a_i = \sum_{i \in \mathcal{N}_- \cap \hat{\mathcal{N}}_+} a_i. \tag{8}$$

Also, for every $\boldsymbol{\sigma}$ with the same composition as $\hat{\boldsymbol{\sigma}}$, $|\mathcal{N}_+ \cap \hat{\mathcal{N}}_-| = |\mathcal{N}_- \cap \hat{\mathcal{N}}_+|$. Let then $n \triangleq |\mathcal{N}_+ \cap \hat{\mathcal{N}}_-| = |\mathcal{N}_- \cap \hat{\mathcal{N}}_+|$. Every $\boldsymbol{\sigma}$ with the same composition as $\hat{\boldsymbol{\sigma}}$ corresponds to a choice of particular subsets ($\mathcal{N}_+ \cap \hat{\mathcal{N}}_-$ and $\mathcal{N}_- \cap \hat{\mathcal{N}}_+$, both of size n) of $\hat{\mathcal{N}}_-$ and $\hat{\mathcal{N}}_+$, respectively. For a given n, the number of combinations of these subsets is the product of the binomial coefficients in the last line of (7). For each such combination, the probability of the event (8) is $\sum_s (\Lambda_s^n / L^n)^2$. The last line of eq. (7) exhausts the product of this probability by the number of combinations for all possible values of n.

So far our analysis has been exact. We now need an evaluation of the exponential order of Λ_s^n, where s scales like nL, i.e., $s = \zeta n L$, $\zeta \in (0,1)$, and we expect the behavior to be symmetric in ζ about the point $\zeta = 1/2$. So it is enough to cover the range $\zeta \in (0, 1/2)$. The event $\sum_{i=1}^n \alpha_i = s$ is obviously equivalent to the event $\sum_{i=1}^n x_i = \zeta n$, where $x_i = \alpha_i / L$. The number of points $\boldsymbol{x} = (x_1, \ldots, x_n) \in \{1/L, 2/L, \ldots, 1 - 1/L, 1\}^n$ with $\sum_{i=1}^n x_i = \zeta n$ is exactly the same as number of points $(x_1, \ldots, x_{n-1}) \in \{1/L, 2/L, \ldots, 1 - 1/L, 1\}^{n-1}$ which satisfy $n\zeta - 1 \leq \sum_{i=1}^{n-1} x_i \leq n\zeta$, which is with excellent approximation given by L^{n-1} times the volume of the region within the unit cube $[0,1]^{n-1}$ of continuous valued $(n-1)$–vectors $(y_1, \ldots, y_{n-1})$ that satisfy $n\zeta - 1 \leq \sum_{i=1}^{n-1} y_i \leq n\zeta$ (see Appendix A for more details). This volume in turn has an exact formula (see, e.g., [7, Theorems 1,4]), which is given in the form of a sum of expressions with alternating signs, but it is not trivial to assess the exponential order of this formula in a compact manner.

Alternatively, we may think of the volume of the set $\{n\zeta - 1 \leq \sum_{i=1}^{n-1} y_i \leq n\zeta\}$ as the probability of the event $\{n\zeta - 1 \leq \sum_{i=1}^{n-1} Y_i \leq n\zeta\}$, where $\{Y_i\}$ are i.i.d. random variables all uniformly distributed over $[0,1]$. Now, denoting the unit step function by $u(x)$ and using the fact that it is the inverse Laplace transform of the function $1/s$, i.e.,

$$u(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \mathrm{d}s \cdot \frac{e^{sx}}{s}, \qquad c > 0,$$

we can represent this probability as the following integral in the complex plane:

$$\begin{aligned}
&\Pr\left\{ n\zeta - 1 \leq \sum_{i=1}^{n-1} Y_i \leq n\zeta \right\} \\
&= \left\langle u\left(n\zeta - \sum_{i=1}^{n-1} Y_i \right) - u\left(n\zeta - 1 - \sum_{i=1}^{n-1} Y_i \right) \right\rangle \\
&= \left\langle \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \mathrm{d}s \exp\left[s\left(n\zeta - \sum_{i=1}^{n-1} Y_i \right) \right] \cdot \frac{(1 - e^{-s})}{s} \right\rangle \\
&= \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \mathrm{d}s e^{s\zeta n} \left\langle \exp\left(-s \sum_{i=1}^{n-1} Y_i \right) \right\rangle \cdot \frac{(1 - e^{-s})}{s}
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \mathrm{d}s e^{s\zeta n}\left\langle e^{-sY_1}\right\rangle^{n-1}\cdot\frac{(1-e^{-s})}{s}\\
&= \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \mathrm{d}s e^{s\zeta n}\left(\frac{1-e^{-s}}{s}\right)^{n-1}\cdot\frac{(1-e^{-s})}{s}\\
&= \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \mathrm{d}s e^{s\zeta n}\left(\frac{1-e^{-s}}{s}\right)^{n}\\
&= \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \mathrm{d}s \exp\left\{n\left[s\zeta+\ln(1-e^{-s})-\ln s\right]\right\}. \qquad (9)
\end{aligned}
$$

This integral is easily evaluated using the saddle point method. In Appendix B, we show that the highest modulus of the integrand along the integration path, which is the vertical straight line $\mathrm{Re}(s)=c$, is uniquely obtained at $s=c$, which yields

$$\Pr\left\{n\zeta-1\le\sum_{i=1}^{n-1}Y_i\le n\zeta\right\}\doteq e^{-n\Phi(\zeta)} \qquad (10)$$

where

$$\Phi(\zeta)=\max_{t\ge 0}[\ln t-\ln(1-e^{-t})-\zeta t],\quad \zeta\in(0,1/2) \qquad (11)$$

and we extend the definition of $\Phi(\cdot)$ to the interval $(0,1)$ to be symmetric around $\zeta=1/2$, namely, $\Phi(1/2-\zeta)=\Phi(\zeta)$. Note that maximizing t is the saddle point of the integral (9), namely, it is the point at which the derivative of the expression in the square brackets vanishes. Also, the axis [4, Section 5.4, p. 84] of this saddle point is in the vertical direction, which is the natural direction of integration path anyway. Thus, we now have

$$\Lambda_s^n\approx L^{n-1}\cdot e^{-n\Phi(\zeta)};\qquad s=\zeta nL,\quad \zeta\in(0,1). \qquad (12)$$

On substituting this into the inner summation of (7), we get

$$
\begin{aligned}
\sum_{s=n}^{nL}(\Lambda_s^n)^2 &\doteq L^{2(n-1)}\sum_{s=n}^{nL}\exp\left\{-n\Phi\left(\frac{s}{nL}\right)\right\}\\
&= nL^{2n-1}\sum_{s=n}^{nL}\frac{1}{nL}\exp\left\{-n\Phi\left(\frac{s}{nL}\right)\right\}\\
&\doteq L^{2n-1}\int_0^1 \mathrm{d}\zeta\exp\{-n\Phi(\zeta)\}\\
&\doteq L^{2n-1}e^{-n\inf_{0<\zeta<1}\Phi(\zeta)}\\
&\doteq L^{2n-1}, \qquad (13)
\end{aligned}
$$

where the last step follows from the fact that the infimum of $\Phi(\zeta)$ over $\zeta\in(0,1)$ is zero (achieved at $\zeta=1/2$). Thus,

$$
\begin{aligned}
\langle\Omega(E(\hat{\boldsymbol{\sigma}}),M(\hat{\boldsymbol{\sigma}}))\rangle &\doteq 1+\frac{1}{L}\sum_n\left(\begin{array}{c}N_+\\ n\end{array}\right)\left(\begin{array}{c}N_-\\ n\end{array}\right)\\
&= 1+\frac{1}{L}\cdot\exp_2\left\{N\sup_{0<\alpha<\min\{p,q\}}\left[ph\left(\frac{\alpha}{p}\right)+qh\left(\frac{\alpha}{q}\right)\right]\right\}\\
&= 1+\frac{1}{L}\cdot 2^{Nh(p)}. \qquad (14)
\end{aligned}
$$

Thus, for $L = 2^{NR}$, we have

$$\langle \Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}})) \rangle - 1 \doteq 2^{N[h(p)-R]}, \tag{15}$$

which means that there is a phase transition at the critical rate of $R_c = h(p)$, as mentioned earlier. For $R > h(p)$, the expression $\langle \Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}})) \rangle - 1$ decays exponentially, and hence so does $\Pr\{\Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}})) > 1\}$, which means that the decoding is unambiguous and correct with high probability. On the other hand, for $R < h(p)$ the probability for the existence of many additional solutions $\{\boldsymbol{\sigma}\}$ must be very high: Since the number of typical source sequences (i.e., sequences with $M(\boldsymbol{\sigma}) = M(\hat{\boldsymbol{\sigma}}) = N(p-q)$) is exponentially $2^{Nh(p)}$ and the number of distinct values of $E(\boldsymbol{\sigma})$ cannot exceed $N \cdot L = N \cdot 2^{NR}$, the fraction of sequences $\{\boldsymbol{\sigma}\}$ (with the given composition) that are *unique* solutions to the equation $E(\boldsymbol{\sigma}) = E(\hat{\boldsymbol{\sigma}})$ cannot exceed $N2^{NR}/2^{Nh(p)} \doteq 2^{-N[h(p)-R]}$, and so, $\Pr\{\Omega(E(\hat{\boldsymbol{\sigma}}), M(\hat{\boldsymbol{\sigma}})) > 1\} \geq 1 - 2^{-N[h(p)-R]}$.

This result is actually quite expected. The critical rate R_c cannot be strictly larger than $h(p)$ because, as said, the total number of sequences with composition (p, q) is exponentially $2^{Nh(p)}$: Had R_c been larger than $h(p)$ we would have obtained that $\left\langle \Omega(\hat{E}) \right\rangle$ grows with an exponential rate which is faster than $h(p)$ (at least when L is subexponential), which is impossible. On the other hand, R_c cannot be strictly smaller than $h(p)$, because then it would mean that Sasmoto's coding scheme achieves a compression ratio that is better than the entropy. Thus, R_c must be equal to $h(p)$.

The above derivation extends straightforwardly in several directions (one at a time or simultaneously):

1. Lossless source coding in the presence of correlated side information at the decoder. Consider the case where the decoder has access to a side information sequence $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_N)$, which is correlated to the source sequence according to a given joint distribution $P(\sigma, \tau)$, and the two sequences are i.i.d. over time, i.e.,

$$P(\boldsymbol{\sigma}, \boldsymbol{\tau}) = \prod_{i=1}^{N} P(\sigma_i, \tau_i). \tag{16}$$

It is well known (see, e.g., [3, Section 15.8]) that in this case, the best achievable compression ratio is given by the conditional entropy of the source given the side information, even if the encoder does not have access to the side information (Slepian–Wolf coding [14]). Here we propose an alternative way to achieve this optimum compression ratio based on subset sum encoding: The encoder works essentially in the same manner as before. The decoder seeks solutions to the equation $\sum_i a_i \sigma_i = E(\hat{\boldsymbol{\sigma}})$ only within the set of $\boldsymbol{\sigma}$–vectors whose joint empirical distribution together with the side information sequence is close to the joint distribution $P(\sigma, \tau)$ (within some small tolerance). In this case, an analysis similar to the above, reveals that the critical rate is given by the conditional entropy of the source given the side information.

2. Multiple Subset Sums. Instead of one set of random variables $a_1, \ldots, a_N$, randomly drawn in $\{1, 2, \ldots, L\}$, consider an array of random variables $\{a_i^k\}$, $i = 1, 2, \ldots, N$, $k = 1, 2, \ldots, m$, all statistically independent, where each a_i^k is uniformly distributed in $\{1, 2, \ldots, L_k\}$, where $L_k = 2^{NR_k}$, $R_k > 0$. The encoder encodes $\boldsymbol{\sigma}$ into $(M(\boldsymbol{\sigma}), E_1(\boldsymbol{\sigma}), \ldots, E_m(\boldsymbol{\sigma}))$, where each $E_k(\sigma) \stackrel{\triangle}{=} \sum_i a_i^k \sigma_i$ is represented by $\log(NL_k) = \log N + NR_k$ bits. The decoder reconstructs $\boldsymbol{\sigma}$ as the first vector whose encoding agrees with the given compressed input $(M(\boldsymbol{\sigma}), E_1(\boldsymbol{\sigma}), \ldots, E_m(\boldsymbol{\sigma}))$. It is easy to show that the decoding is unambiguous with high probability iff $\sum_k R_k > H$. Thus, here the phase transition occurs at the whole hyperplane $\sum_k R_k = H$.

3. General finite alphabets. Another natural direction of extending the above result is from the case of a binary source alphabet to a general finite alphabet which, without loss of generality, will be assumed to be $\Sigma = \{1, 2, \ldots, K\}$. A simple strategy is to decompose the problem into $K - 1$ binary encoding problems, and in each one of them we can rely directly on the binary code construction. Let the input source string $\boldsymbol{\sigma}$ have $N_s = Np_s$ occurrences of $\sigma_i = s \in \Sigma$, $s = 1, 2, \ldots, K$ (of course, $\sum_{s=1}^K N_s = N$). Now, for each $s = 1, 2, \ldots, K-1$, let us select independently at random $a_1^s, a_2^s, \ldots, a_N^s$, each one drawn under the uniform distribution over the integers $\{1, 2, \ldots, L_s\}$, where $L_s = 2^{NR_s}$ and R_s will be specified shortly. The resulting random selections are revealed to the encoder and the decoder. The encoder works as follows: Given the source vector $\boldsymbol{\sigma}$, we first encode the numbers $N_1, N_2, \ldots, N_{K-1}$ as overhead (just like the transmission of $M(\sigma)$ in the binary case). Next, for each $s = 1, \ldots, K-1$, we calculate $E_s(\boldsymbol{\sigma}) = \sum_{i:\ \sigma_i = s} a_i^s - \sum_{i:\ \sigma_i > s} a_i^s$ and transmit its value using $\log(N \cdot L_s)$ bits. The role of each $E_s(\boldsymbol{\sigma})$ is to represent the information pertaining to all locations where $\sigma_i = s$. Based on the results of the binary alphabet case, in order to decode $E_1(\boldsymbol{\sigma})$ unambiguously, R_1 should be at least as large as $h(p_1)$ (think of encoding the binary sequences $\{\mathcal{I}(\sigma_i = 1)\}$, where $\mathcal{I}(\cdot)$ is the indicator function). For the next stage, the task is to fill in N_2 out of the remaining $(N - N_1)$ locations by $s = 2$, and so we have reduced the problem to that of encoding the binary sequence $\{\mathcal{I}(\sigma_i = 2)\}_{i:\ \sigma_i \neq 1}$ of length $(N - N_1)$. By the same reasoning then, to decode $E_2(\boldsymbol{\sigma})$ reliably, R_2 should be at least as large as $(1 - p_1)h(p_2/(1 - p_1))$. This procedure continues until $s = K - 1$, where reliable decoding of $E_{K-1}(s)$ requires $R_{K-1} > (1 - p_1 - \ldots - p_{K-2})h(p_{K-1}/(1 - p_1 - \ldots - p_{K-2}))$. The overall coding rate (neglecting the overhead) is then

$$h(p_1) + (1 - p_1)h\left(\frac{p_2}{1 - p_1}\right) + \ldots + (1 - p_1 - \ldots - p_{K-2})h\left(\frac{p_{K-1}}{1 - p_1 - \ldots - p_{K-2}}\right),$$

which is easily shown (using the chain rule of the entropy) to be identical to the entropy of the source

$$H = -\sum_{s=1}^{K} p_s \log p_s.$$

3. Unconstrained Subset–Sum Coding

Returning to the binary case, suppose next that we wish to make the mapping from $\boldsymbol{\sigma}$ to $E(\boldsymbol{\sigma})$ essentially one–to–one over the *entire* source vector space $\{-1,+1\}^N$, and then there would be no need to specify the composition of $\hat{\boldsymbol{\sigma}}$ to the decoder. This corresponds to the unconstrained subset sum problem. How large should R be now? Here, the analysis is similar but somewhat more involved. The point in presenting the analysis for this case is not quite motivated by the usefulness of the data compression scheme itself, but more about demonstrating the applicability of the analysis method. This time, the derivation is as follows:

$$\begin{aligned}
\Omega(E(\hat{\boldsymbol{\sigma}})) &= \sum_{\boldsymbol{\sigma}} \delta\left(\sum_{j=1}^{N} a_j(\hat{\sigma}_j - \sigma_j)\right) \\
&= \sum_{\boldsymbol{\sigma}} \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \cdot e^{i\omega \sum_{j=1}^{N} a_j(\hat{\sigma}_j - \sigma_i)} \\
&= \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \cdot \sum_{\boldsymbol{\sigma}} \prod_{j=1}^{N} e^{i\omega a_j(\hat{\sigma}_j - \sigma_i)} \\
&= \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \prod_{j=1}^{N} \left[\sum_{\sigma_j} e^{i\omega a_j(\hat{\sigma}_j - \sigma_i)}\right] \\
&= \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \prod_{j=1}^{N} \left(1 + e^{i2\omega a_j \hat{\sigma}_j}\right). \qquad (17)
\end{aligned}$$

Taking now the expectation w.r.t. the randomness of $\{a_j\}$, we readily obtain

$$\langle \Omega(E(\hat{\boldsymbol{\sigma}})) \rangle_{\boldsymbol{a}} = \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \left[1 + \phi(\omega)\right]^{N_+} \cdot \left[1 + \phi^*(\omega)\right]^{N_-}. \qquad (18)$$

Assuming that the binary vector $(\hat{\sigma}_1, \ldots, \hat{\sigma}_N)$ is governed by a binary memoryless source with $p = \Pr\{\hat{\sigma}_i = +1\} = 1 - \Pr\{\hat{\sigma}_i = -1\} = 1 - q$, we next take an ensemble average w.r.t. the randomness of $\{\hat{\sigma}_i\}$, and obtain

$$\begin{aligned}
\langle \Omega(E(\hat{\boldsymbol{\sigma}})) \rangle &= \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \sum_{k=0}^{N} \left(\begin{array}{c} N \\ k \end{array} \right) \left[p(1 + \phi(\omega))\right]^k \cdot \left[q(1 + \phi^*(\omega))\right]^{N-k} \\
&= \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \left[1 + p\phi(\omega) + q\phi^*(\omega)\right]^N \\
&= 1 + \sum_{k=1}^{N} \left(\begin{array}{c} N \\ k \end{array} \right) \cdot \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \left[p\phi(\omega) + q\phi^*(\omega)\right]^k \\
&= 1 + \sum_{k=1}^{N} \left(\begin{array}{c} N \\ k \end{array} \right) \cdot L^{-k} \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \left[p \sum_{\ell=1}^{L} e^{i2\omega\ell} + q \sum_{\ell=1}^{L} e^{-i2\omega\ell}\right]^k \\
&= 1 + \sum_{k=1}^{N} \left(\begin{array}{c} N \\ k \end{array} \right) \cdot L^{-k} \sum_{r=0}^{k} \left(\begin{array}{c} k \\ r \end{array} \right) p^r q^{k-r} \times \\
&\quad \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \left(\sum_{\ell=1}^{L} e^{i2\omega\ell}\right)^r \cdot \left(\sum_{\ell=1}^{L} e^{-i2\omega\ell}\right)^{k-r}
\end{aligned}$$

$$
\begin{aligned}
&= 1 + \sum_{k=1}^{N} \left(\begin{array}{c} N \\ k \end{array} \right) \cdot L^{-k} \sum_{r=0}^{k} \left(\begin{array}{c} k \\ r \end{array} \right) p^r q^{k-r} \times \\
&\quad \int_{-\pi}^{+\pi} \frac{\mathrm{d}\omega}{2\pi} \left(\sum_{\ell=r}^{rL} \Lambda_\ell^r e^{i2\omega\ell} \right) \cdot \left(\sum_{\ell=k-r}^{(k-r)L} \Lambda_\ell^{k-r} e^{-i2\omega\ell} \right) \\
&= 1 + \sum_{k=1}^{N} \left(\begin{array}{c} N \\ k \end{array} \right) \cdot L^{-k} \sum_{r=0}^{k} \left(\begin{array}{c} k \\ r \end{array} \right) p^r q^{k-r} \times \\
&\quad \sum_{\ell=\max\{r,k-r\}}^{L\min\{r,k-r\}} \Lambda_\ell^r \Lambda_\ell^{k-r}. \qquad (19)
\end{aligned}
$$

Now, similarly as before

$$
\begin{aligned}
&\sum_{\ell=\max\{r,k-r\}}^{L\min\{r,k-r\}} \Lambda_\ell^r \Lambda_\ell^{k-r} \\
&\doteq L^{k-2} \sum_{\ell=\max\{r,k-r\}}^{L\min\{r,k-r\}} \exp\left\{ -r\Phi\left(\frac{\ell}{rL}\right) - (k-r)\Phi\left(\frac{\ell}{(k-r)L}\right) \right\} \\
&= L^{k-1} \sum_{\ell=\max\{r,k-r\}}^{L\min\{r,k-r\}} \frac{1}{L} \exp\left\{ -r\Phi\left(\frac{\ell}{rL}\right) - (k-r)\Phi\left(\frac{\ell}{(k-r)L}\right) \right\} \\
&\doteq L^{k-1} \int_0^{\min\{r,k-r\}} \mathrm{d}x \exp\left\{ -r\Phi\left(\frac{x}{r}\right) - (k-r)\Phi\left(\frac{x}{k-r}\right) \right\} \\
&\doteq L^{k-1} e^{-k\psi(\beta)}, \qquad (20)
\end{aligned}
$$

where

$$
\psi(\beta) = \min_{0 \le x \le \min\{\beta, 1-\beta\}} \left[\beta\Phi\left(\frac{x}{\beta}\right) + (1-\beta)\Phi\left(\frac{x}{1-\beta}\right) \right],
$$

where $\beta \stackrel{\triangle}{=} r/k$. Denoting $\alpha = k/N$ and substituting this into eq. (19), we obtain

$$
\begin{aligned}
&\langle \Omega(E(\hat{\boldsymbol{\sigma}})) \rangle - 1 \\
&\doteq \frac{1}{L} \cdot \sum_{k=1}^{N} \left(\begin{array}{c} N \\ k \end{array} \right) \sum_{r=0}^{k} \left(\begin{array}{c} k \\ r \end{array} \right) p^r q^{k-r} e^{-k\psi(\beta)} \\
&\doteq \frac{1}{L} \cdot \exp\left\{ N \max_{\alpha} [h(\alpha) + \alpha \max_{\beta} (h(\beta) + \beta \ln p + (1-\beta)\ln q - \psi(\beta))] \right\} \\
&= \frac{1}{L} \cdot \exp\left\{ N \max_{\alpha} [h(\alpha) - \alpha \min_{\beta} (D(\beta\|p) + \psi(\beta))] \right\} \\
&= \frac{1}{L} \cdot \exp\left\{ N \max_{\alpha} [h(\alpha) - \alpha\xi(p)] \right\} \\
&= \frac{1}{L} \cdot 2^{N \log_2[1 + e^{-\xi(p)}]}, \qquad (21)
\end{aligned}
$$

where we have defined

$$
D(\beta\|p) = \beta \ln \frac{\beta}{p} + (1-\beta) \ln \frac{1-\beta}{1-p} \qquad (22)
$$

and

$$
\xi(p) = \min_{\beta} [D(\beta\|p) + \psi(\beta)]. \qquad (23)
$$

Again, if $L = 2^{NR}$, then the phase transition is now at $R = R_c$, where

$$R_c = \log_2[1 + e^{-\xi(p)}]. \tag{24}$$

Note the special care should be exercised in the extreme cases where $p = 0$ and $p = 1$. It is easy to see that in these cases $D(\beta\|p) = \infty$ for all β, except $\beta = p$ but when $\beta = 0$ and $\beta = 1$, $\psi(\beta) = \infty$, thus the sum $D(\beta\|p) + \psi(\beta)$ is infinite for every $\beta \in [0, 1]$. The choice $\alpha = 0$ is actually not allowed in the out–most maximization over α since the sum over k begins with $k = 1$. Thus, for $p = 0$ and $p = 1$, we have $R_c = -\infty$, which means that $\left\langle \Omega(\hat{E}) \right\rangle = 1$, as expected.

Obviously, R_c cannot be smaller than the entropy of the source.‖ In general, R_c is larger than the entropy, but the coding rate of the corresponding data compression scheme need not be as large as R_c. By applying variable–rate entropy coding to $E(\boldsymbol{\sigma})$, taking advantage of the non–uniform distribution of this random variable, one can compress it at a rate very close to the entropy of the source. While in this case, R_c no longer has the meaning of coding rate, it does another meaning, which is related to the storage requirement for saving the numbers $\{a_i\}$. Note that for $R = 1$, it is easy to specify a particular set of integers $\{a_i\}$ that yields a one–to–one mapping from $\boldsymbol{\sigma}$ to $E(\boldsymbol{\sigma})$: By setting $a_i = 2^{i-1}$, $E(\boldsymbol{\sigma})$ becomes the standard binary representation of $\boldsymbol{\sigma}$ (up to a fixed shift). The above result tells us that we can manage with less storage since R_c is in general less than 1, except the case $p = 1/2$, where $R_c = 1$ (see also [9, Proposition 7.6 and Exercise 7.8]).

Acknowledgment

Useful discussions with Alexander Vardy and with Tuvi Etzion, concerning the evaluation of Λ_s^n (which appears first in eq. (7)), are acknowledged with thanks. In particular, A. Vardy has drawn my attention to ref. [7] in this context and pointed out its relevance.

Appendix

A. Estimating Λ_s^n

Given a lattice and given a certain region in space, the number of lattice sites in that region is approximately given by the volume of the region divided by the volume of a single Voronoi cell pertaining to that lattice. This approximation improves as the ratio between these two volumes becomes very large. More precisely, there is a slight correction due to the fact that some of these Voronoi cells may not be entirely included in the region in question. To obtain upper and lower bounds on the number of lattice points, one may slightly expand (for an upper bound) or shrink (for a lower bound) the given region by an amount that guarantees full inclusion (for an upper lower bound) or

‖ Had it been smaller, we could have compressed at a rate below the entropy using Sasamoto's coding scheme, which is a contradiction [3].

full exclusion (for a lower bound) of the partially included Voronoi cells that are near the boundaries. In our case, we have a cubic lattice in $(n-1)$ dimensions with spacings of $1/L$ in each dimension, so the volume of a Voronoi cell is $1/L^{n-1}$. By replacing $n\zeta$ with $n\zeta \pm n/L$, one can obtain the aforementioned upper and lower bounds, but the corrections of $\pm n/L$ to $n\zeta$ and to $n\zeta - 1$ have negligible effects in the exponential scale since L grows exponentially with n and hence n/L vanishes in the limit.

B. Saddle Point of the Integral (9)

The modulus of the integrand depends solely on the real part of the exponent of the integrand, namely, on $\mathrm{Re}\{s\zeta + \ln(1-e^{-s}) - \ln s\}$. Now, consider an arbitrary complex number $s = r + i\omega$. Then obviously,

$$\mathrm{Re}\{s\zeta + \ln(1-e^{-s}) - \ln s\} = r\zeta + \frac{1}{2}\ln\left(\frac{1+e^{-2r}-2e^{-r}\cos\omega}{r^2+\omega^2}\right). \tag{25}$$

Thus, we have to show that

$$\frac{1+e^{-2r}-2e^{-r}\cos\omega}{r^2+\omega^2}$$

is maximized at $\omega = 0$, and only at $\omega = 0$, for all $r > 0$. This would be equivalent to the assertion that

$$\frac{(1-e^{-r})^2}{r^2} \geq \frac{1+e^{-2r}-2e^{-r}\cos\omega}{r^2+\omega^2}, \tag{26}$$

with equality if and only if $\omega = 0$, since the left–hand side is obtained from the right–hand side by setting $\omega = 0$.

In order to prove eq. (26), we begin from the obvious inequality $\theta \geq \sin(\theta)$, which holds for all $\theta \geq 0$. Integrating both sides of this inequality from 0 to ω ($\omega > 0$), we obtain $\omega^2/2 \geq 1 - \cos(\omega)$ with equality if and only if $\omega = 0$, or equivalently,

$$\frac{\omega^2}{1-\cos(\omega)} \geq 2, \tag{27}$$

which now holds for all ω since the left–hand side is an even function. Similarly, beginning from the inequality $1 \leq \cosh(r)$ and integrating both sides twice, we get $r^2/2 \leq \cosh(r) - 1$, or equivalently

$$\frac{r^2}{\cosh(r)-1} \leq 2, \tag{28}$$

which when combined with (27) yields

$$\frac{r^2}{\cosh(r)-1} \leq \frac{\omega^2}{1-\cos(\omega)}, \tag{29}$$

or equivalently,

$$\frac{2r^2e^{-r}}{(1-e^{-r})^2} \leq \frac{\omega^2}{1-\cos(\omega)}, \tag{30}$$

which is

$$2r^2e^{-r} - 2r^2e^{-r}\cos(\omega) \leq \omega^2(1-e^{-r})^2. \tag{31}$$

Adding $r^2(1-e^{-r})^2$ to both sides of this inequality and rearranging terms, one obtains

$$r^2[1+e^{-2r}-2e^{-r}\cos(\omega)] \leq (r^2+\omega^2)(1-e^{-r})^2, \tag{32}$$

which is equivalent to eq. (26).

References

[1] C. Borgs, J. Chayez, and B. Pittel, *Random Struct. Algorithms*, **19** 247-288, 2001.

[2] C. Borgs, J. Chayez, S. Mertens, and B. Pittel, *Random Struct. Algorithms*, **24** 315–380, 2003.

[3] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, second edition, John Wiley & Sons, 2006.

[4] N. G. de Bruijn, *Asymptotic Methods in Analysis*, Dover Publications, New York, 1981.

[5] F. F. Ferreira and J. F. Fontanari, *J. Phys. A*, **31**, 3417–3428, 1998.

[6] M. R. Garey and D. S. Johnson, *Computers and Intractability: A Guide to the Theory of NP–Completeness*, W. H. Freeman, New York , 1979.

[7] J.-L. Marichal and M. J. Mossinghoff, *Online Journal of Analytic Combinatorics*, Issue 3, 2008. https://sites.google.com/site/analyticcombinatorics/archives-2

[8] S. Mertens, *Theoretical Computer Science*, **265** 79–108, 2001.

[9] M. Mézard and A. Montanari, *Information, Physics and Computation*, Oxford University Press, 2009.

[10] H. Nishimori, *Statistical Physics of Spin Glasses and Information Processing: an Introduction*, (International Series of Monographs on Physics, no. 111), Oxford University Press, 2001.

[11] C. H. Papadimitriuou, *Computational Complexity*, Addison–Wesley, Reading, MA, 1994.

[12] T. Sasamoto, *Physica A*, **321**, 369–374, 2003.

[13] T. Sasamoto, T. Toyoizumi, and H. Nishimori, *J. Phys. Math. Gen.*, **34**, 9555–9567, 2001.

[14] D. Slepian and J. K. Wolf, *IEEE Trans. Inform. Theory*, **IT–19**, 471–480, 1973.